

Aditya Kulkarni

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PROFESSIONAL SUMMARY

Full stack engineer with production Python and TypeScript -- building ML fine-tuning pipelines (sentence-transformers, in-batch negatives, MS MARCO), multi-model LLM tooling (Claude, GPT-4o, Gemini + MCP), and TypeScript/React/Node.js backends at Isomer AI across 394 commits and 7 production deployments. Designed Kafka-driven data pipelines, REST APIs with clean ergonomics, and observability stacks (Prometheus, Grafana, Sentry) from scratch. Comfortable owning the full stack -- backend services, developer-facing APIs, and product console -- in fast-moving, ambiguous environments.

CORE COMPETENCIES

Python + FastAPI

TypeScript + Node.js

React + Next.js

ML Fine-Tuning + Inference

LLM + Multi-Model AI

Kafka + Distributed Systems

Prometheus + Grafana + Sentry

REST API Design + DX

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern - Raleigh, NC

- Built multi-model LLM platform integrating **Claude, GPT-4o, and Gemini** (8+ models, 3 providers) via MCP; implemented Temporal.io workflow orchestration and real-time event pipelines as core agentic system components -- TypeScript, Node.js, Express, PostgreSQL, Redis, Prisma, Docker
- Shipped 10+ backend features across **394 commits and 7 production deployments**; designed composable REST APIs with Auth0 RBAC, RFC 9457 error responses, and clean ergonomics reviewed by non-engineer team members; instrumented production with Sentry
- Built document processing pipelines (PDF generation, Microsoft Graph API email parsing, S3 file storage) and developer-facing tooling supporting cross-functional workflows

NC State University Apr 2025 - Present

Graduate Research Assistant - Raleigh, NC

- Built a GCP Cloud Run platform enabling 100+ researchers to interactively explore 500GB genomic datasets; applied hyperparameter tuning and time-series analysis, improving ML model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ AI-driven workflows including a health claim automation tool (OCR + NLP + ML); improved accuracy by 25%; automated AWS and GCP deployments with Grafana monitoring

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Implemented distributed data pipelines (Hadoop + Spark + Kafka) for real-time ticket ingestion, reducing latency by 20%; led full SDLC reducing processing time by 15%

PROJECTS

VectorLift -- Semantic Search + ML Fine-Tuning Python + PyTorch + Kafka github.com/Aditya-k24/VectorLift

Fine-tuned sentence-transformers with in-batch negatives on MS MARCO for dense bi-encoder retrieval; cross-encoder re-ranker (MiniLM) as re-ranking stage; evaluated across 6 pipeline configurations with NDCG@10, MRR@10, MAP, and paired bootstrap significance testing. Kafka-driven indexing pipeline for event-based corpus ingestion into Elasticsearch and Qdrant; FastAPI search server with per-stage latency tracking; Streamlit experiment dashboard; full observability via Prometheus and Grafana.

Python, PyTorch, Sentence-Transformers, Kafka, Elasticsearch, Qdrant, FastAPI, Streamlit, PostgreSQL, Redis, Prometheus, Grafana, Docker

LLM Eval Harness -- Multi-Model Benchmark

Claude + GPT-4o + Gemini

github.com/Aditya-k24/llm-eval-harness

Vendor-neutral benchmark comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 3 task types (grounded QA, multi-hop QA, FEVER claim verification) on a 2,000-example frozen test set with 3 repeated runs per model. Metrics: ExactMatch, TokenF1, abstention accuracy, grounded hallucination rate, p50/p95 latency, cost/call; statistical significance via paired bootstrap CIs, McNemar test, and Holm correction.

Python, Anthropic/OpenAI/Google GenAI SDKs, DuckDB, Parquet, Streamlit, Docker

AlgoPulse -- Agentic AI Orchestration Platform

TypeScript + Temporal + Kafka

github.com/Aditya-k24/AlgoPulse

Event-driven agentic AI platform scaled to 2,000 async inference requests/second with Temporal.io and Kafka for state synchronization and fault tolerance; identified API-polling bottleneck under load and pivoted architecture to Supabase Edge Functions for streaming AI reasoning. TypeScript, Node.js, React front-end console.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

PulseRank -- Real-Time Leaderboard Platform

TypeScript + NestJS

github.com/Aditya-k24/PulseRank

Event-driven leaderboard and activity feed: MySQL transactional writes with idempotency-key deduplication and outbox pattern, MongoDB feed projections with TTL expiry, Redis sorted sets for sub-millisecond leaderboard reads, live updates over Server-Sent Events. OpenTelemetry + Prometheus observability; GitHub Actions CI/CD.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus, GitHub Actions

EDUCATION

North Carolina State University -- MS Computer Science

Aug 2024 - May 2026

Raleigh, NC | Courses: Artificial Intelligence, Operating Systems, Software Engineering, Algorithms, Data Science

Mumbai University -- BTech Computer Science

Aug 2020 - May 2024

CGPA: 9.32/10 | Courses: Data Structures & Algorithms, Machine Learning, Software Engineering, Big Data Infrastructure

SKILLS

Languages: Python, TypeScript, Java, JavaScript, SQL

ML/AI: PyTorch, Sentence-Transformers, fine-tuning, RAG, Claude/OpenAI/Gemini APIs, MCP, LangChain, Hugging Face

Backend: FastAPI, Node.js, Express, NestJS, REST APIs, Kafka, Temporal.io, PostgreSQL, Redis, MongoDB, Docker

Frontend: React, Next.js, Streamlit, Tailwind CSS **Observability:** Prometheus, Grafana, Sentry, OpenTelemetry

Cloud: AWS (S3, automation), GCP (Cloud Run), GitHub Actions CI/CD